From Δ(ε) to Δ(E)

The gap function of the Rashba + correlated-hopping model as a function of the quasiparticle energy — step-by-step derivation, including the band test that discards unphysical roots

Paper: Joel Hutchinson, J. E. Hirsch, Frank Marsiglio, *Enhancement of Superconducting T_c due to the Spin-orbit Interaction*, Phys. Rev. B **97**, 184513 (2018), arXiv:1801.08988. "Paper Eq. n" always refers to that article; the equations of this note are numbered (1), (2), …

Result. Let $b \equiv \Delta^s/4t$, $\Delta_F \equiv \Delta^0 - \mu b$, $E_{\min} \equiv |\Delta_F|/\sqrt{1+b^2}$, $\varepsilon^* \equiv (\mu + b\Delta^0)/(1+b^2)$ and $W \equiv 4t\sqrt{1+V_{SO}^2/2t^2}$ (half-width of the Rashba band). Solving $E(\varepsilon) = E$ gives two *mathematical* branches, "+" with $\varepsilon > \varepsilon^*$ and "−" with $\varepsilon < \varepsilon^*$, defined for $E \geq E_{\min}$:

$$\Delta_{\pm}(E) = \frac{\Delta_F}{1+b^2} \mp \frac{b}{\sqrt{1+b^2}}\sqrt{E^2 - E_{\min}^2} \tag{R}$$

$$\varepsilon_{\pm}(E) = \varepsilon^* \pm \frac{\sqrt{E^2 - E_{\min}^2}}{\sqrt{1+b^2}} \tag{R'}$$

Only branches whose band energy lies inside the Rashba band are real states. The test, applied to each sign separately, is

$$-W \leq \varepsilon_{\pm}(E) \leq +W \tag{A}$$

A branch with $|\varepsilon_{\pm}(E)| > W$ is discarded: no $(\mathbf{k}, s)$ state has quasiparticle energy E on it, so $\Delta_{\pm}(E)$ does not exist there. For ε^* inside the band, $\Delta(E)$ is therefore two-valued only for $E_{\min} \leq E \leq \min(E_{+W}, E_{-W})$, single-valued up to $\max(E_{+W}, E_{-W})$ and undefined above that, where $E_{\pm W} \equiv E(\varepsilon = \pm W)$ is the quasiparticle energy at the band edge (Sec. 3; Sec. 3.3 gives the case of ε^* outside the band).

0. Which energy is which

Four energies appear:

symbol	meaning	where
$\varepsilon \equiv \varepsilon_{\mathbf{k}s} = \varepsilon_{\mathbf{k}} - 2sV_{SO}R_{\mathbf{k}}$	normal-state Rashba band energy, $s = \pm 1$ helicity, $R_{\mathbf{k}} \equiv \sqrt{\sin^2 k_x + \sin^2 k_y}$	paper Eq. 15
$\xi \equiv \varepsilon - \mu$	band energy measured from the chemical potential	–
$E \equiv E_{\mathbf{k}s} = \sqrt{(\varepsilon_{\mathbf{k}s} - \mu)^2 + \lvert\Delta_{\mathbf{k}s}\rvert^2}$	Bogoliubov quasiparticle energy: the argument of the Fermi function in the gap equation (paper Eq. 30) and the energy a tunnelling bias measures ($eV = E$)	paper Eq. 26
$W \equiv 4t\sqrt{1+V_{SO}^2/2t^2}$	half-width of the Rashba band: every state has $-W \leq \varepsilon_{\mathbf{k}s} \leq W$	Sec. 3

The paper's result is $\Delta(\varepsilon) = \Delta^0 - \Delta^s \varepsilon/4t$ (paper Eq. 37). **"$\Delta(E)$" is taken to mean the gap of the state whose quasiparticle energy is E.** If you instead meant $E \equiv \varepsilon - \mu$, nothing needs inverting: (6) below is already $\Delta(E) = \Delta_F - bE$ (restricted, as always, to the band, $-W \le E + \mu \le W$).

Notation clashes in the paper: $g(E)$ is the density of states in paper Eq. 18 (there E is a *band* energy) and $g(E) \equiv (1 - 2f(E))/2E$ after paper Eq. 34 (there E is the *quasiparticle* energy); $E_{\min}$ is the bottom of the free Rashba band in Sec. II and the minimum excitation energy in paper Eq. 40. Here $N(\varepsilon)$ is the normal-state density of states ($\int N\, d\varepsilon = 2$, both helicities), the band bottom is $-W$, and $E_{\min}$ is the minimum excitation energy of paper Eq. 40.

1. Starting point: why the gap depends on (k, s) only through ε

The paper's ansatz (paper Eq. 31), with $s_{\mathbf{k}} = \frac{1}{2}(\cos k_x + \cos k_y)$, is

$$\bar{\Delta}_{\mathbf{k}s} = \Delta^0 + \Delta^s s_{\mathbf{k}} + \Delta^{x-y}\, s\, R_{\mathbf{k}} \tag{1}$$

Paper Eqs. 33 and 34 have the same $\mathbf{k}'$-sum on the right-hand side, so their ratio gives (paper Eq. 35)

$$\frac{\Delta^{x-y}}{\Delta^s} = \frac{4(\Delta t/t)\, V_{SO}}{8\,\Delta t} = \frac{V_{SO}}{2t} \tag{2}$$

Now use $\varepsilon_{\mathbf{k}} = -2t(\cos k_x + \cos k_y)$, i.e. $s_{\mathbf{k}} = -\varepsilon_{\mathbf{k}}/4t$, and paper Eq. 15, i.e. $sV_{SO}R_{\mathbf{k}} = -(\varepsilon_{\mathbf{k}s} - \varepsilon_{\mathbf{k}})/2$:

$$s_{\mathbf{k}} + \frac{sV_{SO}}{2t} R_{\mathbf{k}} = -\frac{\varepsilon_{\mathbf{k}}}{4t} - \frac{\varepsilon_{\mathbf{k}s} - \varepsilon_{\mathbf{k}}}{4t} = -\frac{\varepsilon_{\mathbf{k}s}}{4t} \tag{3}$$

Inserting (2) and (3) into (1) gives paper Eq. 37:

$$\bar{\Delta}_{\mathbf{k}s} = \Delta^0 - \frac{\Delta^s}{4t}\, \varepsilon_{\mathbf{k}s} \tag{4}$$

Since $\Delta_{\mathbf{k}s} = s\, e^{-i\theta_{\mathbf{k}}} \bar{\Delta}_{\mathbf{k}s}$ (paper, text after Eq. 22) differs from $\bar{\Delta}_{\mathbf{k}s}$ only by a phase, $|\Delta_{\mathbf{k}s}| = |\bar{\Delta}_{\mathbf{k}s}|$ and the quasiparticle energy depends on $(\mathbf{k}, s)$ *only through* $\varepsilon_{\mathbf{k}s}$:

$$E(\varepsilon) = \sqrt{(\varepsilon - \mu)^2 + \left(\Delta^0 - \frac{\Delta^s}{4t}\,\varepsilon\right)^2} \tag{5}$$

All dependence on $\mathbf{k}$, s and V_{SO} therefore sits in $\varepsilon_{\mathbf{k}s}$, in the density of states $N(\varepsilon)$, in the band edges $\pm W$ (Sec. 3) and in the self-consistent numbers Δ^0, Δ^s, μ. The inversion $\varepsilon \to E$ is a one-variable problem.

2. The inversion: two mathematical roots (Steps 1–5)

In Steps 1–5, ε is treated as an arbitrary real number. They give the two *mathematical* solutions of $E(\varepsilon) = E$; **Step 6 (Sec. 3) then decides which of them are physical.**

Step 1 — measure energies from the Fermi level.

Put $\xi = \varepsilon - \mu$, $b \equiv \Delta^s/4t$ and $\Delta_F \equiv \Delta(\varepsilon = \mu) = \Delta^0 - b\mu$. Then

$$\Delta(\xi) = \Delta^0 - b(\mu + \xi) = \Delta_F - b\,\xi \tag{6}$$

Step 2 — write E^2 as a polynomial in ξ.

$$E^2 = \xi^2 + (\Delta_F - b\xi)^2 = (1 + b^2)\,\xi^2 - 2b\Delta_F \xi + \Delta_F^2 \tag{7}$$

This is a convex parabola in ξ (leading coefficient $1 + b^2 > 0$): E has exactly one minimum, and every larger E is reached at two values of ξ.

Step 3 — complete the square.

$$E^2 = E_{\min}^2 + (1 + b^2)(\xi - \xi^*)^2, \qquad \xi^* = \frac{b\Delta_F}{1 + b^2}, \quad E_{\min} = \frac{|\Delta_F|}{\sqrt{1 + b^2}} \tag{8}$$

Check: $(1+b^2)(\xi-\xi^*)^2 = (1+b^2)\xi^2 - 2b\Delta_F\xi + b^2\Delta_F^2/(1+b^2)$, and adding $E_{\min}^2 = \Delta_F^2/(1+b^2)$ gives the constant Δ_F^2, as in (7).

In the paper's variables, $\varepsilon^* = \mu + \xi^*$ and $E_{\min}$ read

$$\varepsilon^* = \frac{\mu + \Delta^0\,\Delta^s/4t}{1+(\Delta^s/4t)^2}, \qquad E_{\min} = \frac{\left|\Delta^0 - \mu\,\Delta^s/4t\right|}{\sqrt{1+(\Delta^s/4t)^2}} \tag{9}$$

which are exactly paper Eqs. 39 and 40: a first consistency check. Because $\xi^* \neq 0$ the minimum of $E(\varepsilon)$ is *not* at the Fermi level.

Step 4 — solve for ξ.

$$\xi_\pm(E) = \xi^* \pm \frac{\sqrt{E^2 - E_{\min}^2}}{\sqrt{1+b^2}} = \frac{b\Delta_F \pm \sqrt{(1+b^2)E^2 - \Delta_F^2}}{1+b^2}, \qquad E \geq E_{\min} \tag{10}$$

There is no real root for $E < E_{\min}$. The two roots, $\varepsilon_\pm = \mu + \xi_\pm$, lie above and below ε^*. **They are only mathematical roots: whether a root corresponds to real states is decided in Step 6.**

Step 5 — insert $\xi_\pm(E)$ into $\Delta(\xi) = \Delta_F - b\xi$.

$$\Delta_\pm(E) = \Delta_F - b\,\xi_\pm(E) = \Delta_F - \frac{b^2\Delta_F}{1+b^2} \mp \frac{b\sqrt{(1+b^2)E^2 - \Delta_F^2}}{1+b^2} \tag{11}$$

Using $\Delta_F - b^2\Delta_F/(1+b^2) = \Delta_F/(1+b^2)$ and $(1+b^2)E^2 - \Delta_F^2 = (1+b^2)(E^2 - E_{\min}^2)$ this simplifies to

$$\Delta_\pm(E) = \frac{\Delta_F}{1+b^2} \mp \frac{b}{\sqrt{1+b^2}}\sqrt{E^2 - E_{\min}^2}, \qquad E \geq E_{\min} \tag{R}$$

$$\varepsilon_\pm(E) = \mu + \xi_\pm(E) = \varepsilon^* \pm \frac{\sqrt{E^2 - E_{\min}^2}}{\sqrt{1+b^2}} \tag{R'}$$

Written out with the paper's Δ^0, Δ^s, μ:

$$\Delta_\pm(E) = \frac{\Delta^0 - \mu\frac{\Delta^s}{4t} \mp \frac{\Delta^s}{4t}\sqrt{\left[1+\left(\frac{\Delta^s}{4t}\right)^2\right]E^2 - \left(\Delta^0 - \mu\frac{\Delta^s}{4t}\right)^2}}{1+\left(\frac{\Delta^s}{4t}\right)^2} \tag{12}$$

Equivalent form with $b = \tan\varphi$: $\Delta_\pm(E) = \Delta_F\cos^2\varphi \mp \sin\varphi\sqrt{E^2 - \Delta_F^2\cos^2\varphi}$.

Geometric picture

In the (ξ, Δ) plane, (6) is a straight line and $E = \sqrt{\xi^2 + \Delta^2}$ is the distance from the origin (Fig. 1b). $E_{\min} = |\Delta_F|/\sqrt{1+b^2}$ is the perpendicular distance from the origin to the line, (ξ^*, Δ^*) is the foot of the perpendicular, and a circle of radius $E > E_{\min}$ cuts the line in two points — the two mathematical branches. Eq. (R) says: walk a distance $\sqrt{E^2 - E_{\min}^2}$ along the line, in either direction, from the foot of the perpendicular. Only the part of the line that lies inside the band is physical (solid in Fig. 1b); intersections on the dotted part are the discarded roots.

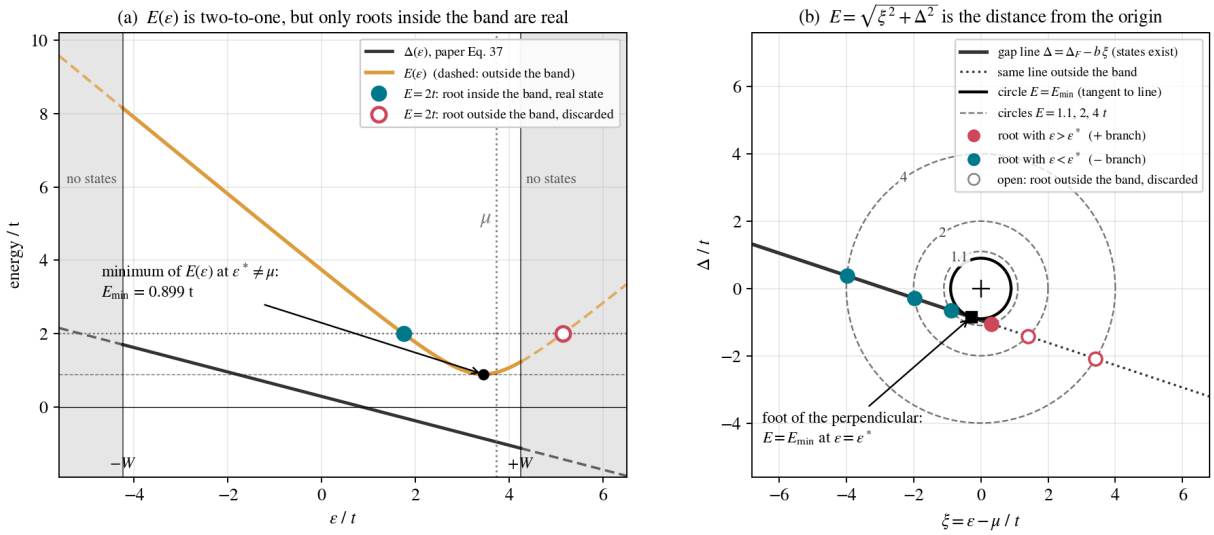


Fig. 1. (a) The gap $\Delta(\varepsilon)$ is linear, but the quasiparticle energy $E(\varepsilon)$ has a minimum at $\varepsilon^* \neq \mu$, so each E is reached at two values of ε. The shaded regions $|\varepsilon| > W$ contain no states: of the two roots at $E = 2t$ only the filled one is a real state, the open one is discarded (Step 6). (b) The same relation in the (ξ, Δ) plane: a circle of radius E meets the gap line at two points (filled: root inside the band; open: root outside the band, discarded). Parameters as in Fig. 3.

3. Step 6 — keep only the roots that are real states (band test)

A root $\varepsilon_{\pm}(E)$ describes a physical quasiparticle only if some $(\mathbf{k}, s)$ actually has that band energy, i.e. only if $\varepsilon_{\pm}(E)$ lies inside the Rashba band. This single question decides which branches survive.

3.1 The allowed band

$$-W \le \varepsilon_{\mathbf{k}s} \le +W, \qquad W = 4t\sqrt{1+\frac{V_{SO}^2}{2t^2}} \tag{13}$$

Proof. For $s=-1$, $\varepsilon_{\mathbf{k}s} = -2t(\cos k_x + \cos k_y) + 2V_{SO}R_{\mathbf{k}}$. With $u = \frac{1}{2}(\cos k_x + \cos k_y) \in [-1,1]$ one has $R_{\mathbf{k}}^2 = 2 - \cos^2 k_x - \cos^2 k_y \le 2(1-u^2)$, hence, by the Cauchy–Schwarz inequality,

$$\varepsilon_{\mathbf{k}s} \le -4tu + 2\sqrt{2}\ V_{SO}\sqrt{1-u^2}\ \le \sqrt{(4t)^2 + 8V_{SO}^2} = W$$

Equality holds on the diagonals $|k_x| = |k_y|$, so the maximum of $\varepsilon_{\mathbf{k}s}$ is exactly W; the $s=+1$ band is the mirror image, with minimum $-W$. Separately, the helicity bands are $s=+1$: $[-W,\ 4t]$ and $s=-1$: $[-4t,\ W]$ (the points (π,π) and $(0,0)$ give the $\pm 4t$ ends). $\varepsilon_{\mathbf{k}s}$ is continuous on the Brillouin zone, so every energy in $[-W, W]$ is realised and none outside. Because $\bar{\Delta}$ and E depend on $(\mathbf{k}, s)$ only through $\varepsilon_{\mathbf{k}s}$ (Sec. 1), the helicity label plays no role in the test below.

3.2 The admissibility rule

$$-W \le \varepsilon_{\sigma}(E) \le +W, \qquad \varepsilon_{\sigma}(E) = \varepsilon^* + \sigma\frac{\sqrt{E^2 - E_{\min}^2}}{\sqrt{1+b^2}}, \quad \sigma = \pm 1 \tag{A}$$

Discard rule. If $|\varepsilon_{\sigma}(E)| > W$, no $(\mathbf{k}, s)$ state has $E_{\mathbf{k}s} = E$ with $\varepsilon_{\mathbf{k}s} = \varepsilon_{\sigma}(E)$. That root is an artefact of continuing the parabola (7) to band energies that do not exist; it is discarded and $\Delta_{\sigma}(E)$ is *undefined* at that E (not zero, not small). It must not enter any plot, average, sum or integral. In Fig. 1 these are the open circles in (b) and the dashed parts of the curves in (a); in Figs. 2 and 3 the dashed lines.

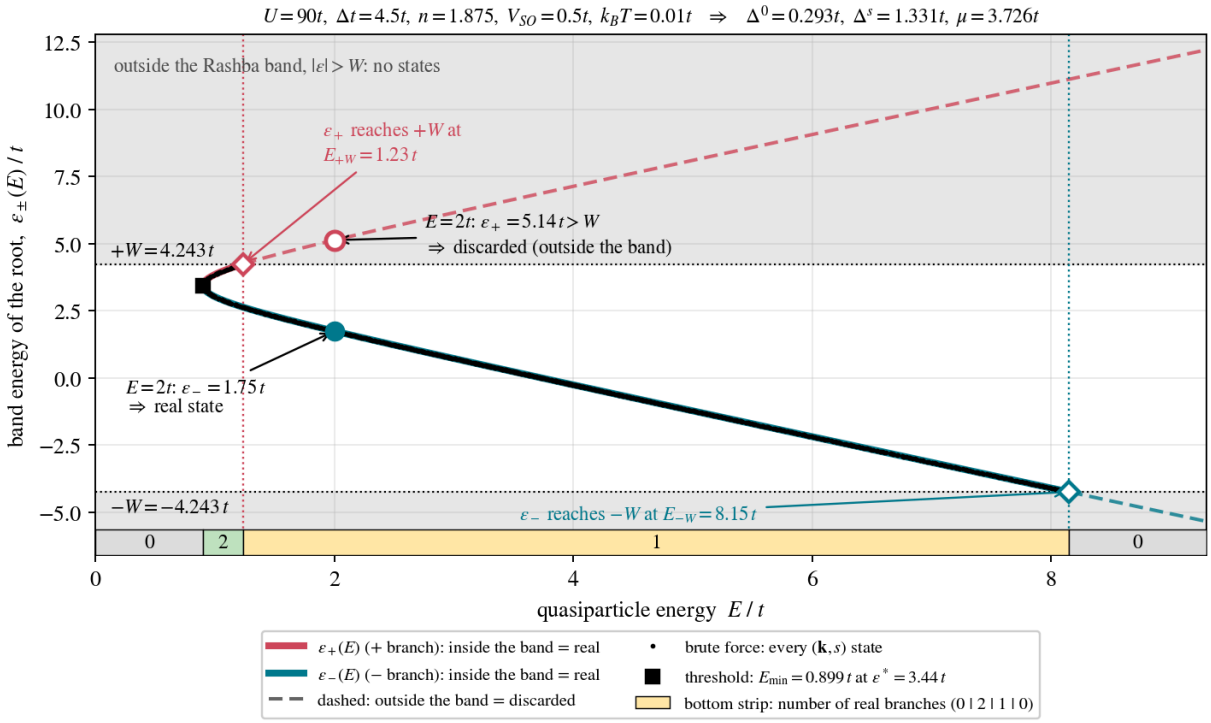


Fig. 2. The band test. Band energies of the two roots, $\varepsilon_\pm(E)$, against E. Solid: root inside the Rashba band (real states). Dashed: root outside the band (discarded). Black dots: $\varepsilon_{\mathbf{k}s}$ against $E_{\mathbf{k}s}$ for every $(\mathbf{k},s)$ state — they lie only on the solid parts. The bottom strip gives the number of real branches: 0 below $E_{\min}$, 2 up to E_{+W}, 1 up to E_{-W}, 0 above.

3.3 The rule written as a condition on E

By (8), E grows monotonically with $|\varepsilon-\varepsilon^*|$. A root therefore reaches a band edge exactly when E equals the quasiparticle energy at that edge,

$$E_{\pm W}\equiv E(\varepsilon=\pm W)=\sqrt{E_{\min}^2+(1+b^2)(\pm W-\varepsilon^*)^2}=\sqrt{(\pm W-\mu)^2+(\Delta^0\mp bW)^2} \tag{14}$$

For ε^* inside the band ($|\varepsilon^*|\le W$, the usual case) the test (A) is therefore equivalent to a simple energy window for each branch:

branch	band energy of the root	real (passes (A)) for
$+$	$\varepsilon_+\ge\varepsilon^*$	$E_{\min}\le E\le E_{+W}$
$-$	$\varepsilon_-\le\varepsilon^*$	$E_{\min}\le E\le E_{-W}$

Consequently: (i) $\Delta(E)$ is **two-valued only** for $E_{\min}\le E\le\ \min\,(E_{+W},E_{-W})$; (ii) for $\min\,(E_{+W},E_{-W})<E\le\ \max\,(E_{+W},E_{-W})$ exactly one branch survives and $\Delta(E)$ is **single-valued**; (iii) above $E_{\max}\equiv\ \max\,(E_{+W},E_{-W})$ there are no states and $\Delta(E)$ does not exist ($E_{\max}$ is the largest quasiparticle energy in the whole zone).

If ε^* lies outside the band (e.g. μ near a band edge together with large b), the $E_{\min}$ of paper Eq. 40 is not reached by any state and the physical threshold moves to a band-edge energy. The general windows are

branch	real for	never real if
$+$	$E(\ \max\,(\varepsilon^*,-W))\le E\le E_{+W}$	$\varepsilon^*>W$
$-$	$E(\ \min\,(\varepsilon^*,W))\le E\le E_{-W}$	$\varepsilon^*<-W$

with $E(\varepsilon)$ from (5); for $|\varepsilon^*|\le W$ they reduce to the previous table.

4. Checks and properties

- **BCS limit** $b \to 0$ (energy-independent gap): $\Delta_\pm(E) \to \Delta_F$, $E_{\min} \to |\Delta_F|$, $\varepsilon_\pm \to \mu \pm \sqrt{E^2 - \Delta_F^2}$.
- **Threshold** $E = E_{\min}$ (for $|\varepsilon^*| \le W$): the root vanishes, both branches meet at $\Delta^* = \Delta_F/(1+b^2) = \Delta(\varepsilon^*)$.
- **High energy**: $\Delta_\pm(E) \to \Delta^* \mp \frac{b}{\sqrt{1+b^2}} E$ (there $E \approx |\xi|\sqrt{1+b^2}$): each formula branch grows linearly with E — within the band, up to $E_{\pm W}$.
- **Branch splitting** (particle–hole asymmetry), where both branches are real: $\Delta_-(E) - \Delta_+(E) = \frac{2b}{\sqrt{1+b^2}}\sqrt{E^2 - E_{\min}^2}$, zero for $b = 0$.
- **Back-substitution**: $\xi_\pm^2 + \Delta_\pm^2 = E^2$ identically (checked symbolically with sympy, and numerically for every $(\mathbf{k}, s)$ state, Fig. 3).
- $E_{\min} = 0$ only if $\Delta_F = 0$, i.e. if the node $\varepsilon_0 = 4t\Delta^0/\Delta^s$ of the gap function lies exactly at μ.
- **Band test, numerically.** At $V_{SO} = 0.5t$ the helicity bands are $s = +1$: [−4.2426, 4.0000] t and $s = -1$: [−4.0000, 4.2426] t, so $W = 4.2426\,t$. Over all 28,800 states, branch $+$ occupies exactly $E \in$ [0.8991, 1.2327] t and branch $-$ exactly [0.8991, 8.1494] t: the windows of the table above ($E_{\min} = 0.8991\,t$, $E_{+W} = 1.233\,t$, $E_{-W} = 8.149\,t$); no state sits on a discarded root. Stress test: 300 random sets of $(\Delta^0, \Delta^s, \mu, V_{SO})$, 35 of them with ε^* outside the band, brute force on a 300×300×2 grid — no state lies outside the predicted window of its branch, and the upper end of every window (branch reaching a band edge) is filled to within $2{\times}10^{-4}\ t$ (grid resolution).

5. Recipe for computing Δ(E)

1. Get Δ^0, Δ^s, μ for your $T, U, \Delta t, n, V_{SO}$ from the self-consistent solution of paper Eqs. 32, 33, 38 ($\Delta^{x-y} = V_{SO}\Delta^s/2t$ then follows). T, U, Δt and n enter only through these three numbers; V_{SO} also enters W.
2. Compute $b = \Delta^s/4t$, $\Delta_F = \Delta^0 - \mu b$, $E_{\min} = |\Delta_F|/\sqrt{1+b^2}$, $\varepsilon^* = (\mu + b\Delta^0)/(1+b^2)$, $\Delta^* = \Delta_F/(1+b^2)$ and $W = 4t\sqrt{1 + V_{SO}^2/2t^2}$.
3. For a given E: if $E < E_{\min}$ there is no state (and none if $E > E_{\max}$).
4. Otherwise put $\ell = \sqrt{E^2 - E_{\min}^2}$, $\varepsilon_\pm = \varepsilon^* \pm \ell/\sqrt{1+b^2}$ and $\Delta_\pm = \Delta^* \mp b\ell/\sqrt{1+b^2}$ — the two mathematical branches.
5. **Band test:** keep a branch only if $|\varepsilon_\pm| \le W$; **discard every branch with** $|\varepsilon_\pm| > W$.
6. What remains (zero, one or two values) is $\Delta(E)$. It keeps the sign of $\bar{\Delta}$: it vanishes at $\varepsilon_0 = 4t\Delta^0/\Delta^s$, i.e. at $E_0 = |\varepsilon_0 - \mu|$ on the branch that contains ε_0.

6. By-products you will probably need next

Differentiating (10) gives the Jacobian, which turns a $\mathbf{k}$-sum into an energy integral. With $S(E) = \{\sigma\colon |\varepsilon_\sigma(E)| \le W\}$, the set of branches that pass the band test (A):

$$\left|\frac{d\varepsilon_\pm}{dE}\right| = \frac{E}{\sqrt{(1+b^2)E^2 - \Delta_F^2}} = \frac{E}{\sqrt{1+b^2}\,\sqrt{E^2 - E_{\min}^2}} \tag{15}$$

so the quasiparticle density of states has a BCS-like inverse-square-root edge at $E_{\min}$, and the sum runs *only over admissible branches*:

$$N_s(E) = \sum_{\sigma \in S(E)} N\left(\varepsilon_\sigma(E)\right) \left|\frac{d\varepsilon_\sigma}{dE}\right| \tag{16}$$

With the BCS coherence factors $v^2 = \frac{1}{2}(1 - \xi/E)$ (paper Eq. 24) and $u^2 = 1 - v^2$, the $T = 0$ tunnelling density of states (momentum-independent matrix element, spin summed; $\eta = +1$ adds an electron, weight u^2, $\eta = -1$ removes one, weight v^2) is

$$N_t(\eta E) = \sum_{\sigma \in S(E)} N\left(\varepsilon_\sigma(E)\right) \frac{E + \eta\,\xi_\sigma(E)}{2\sqrt{(1+b^2)E^2 - \Delta_F^2}} \tag{17}$$

$N(\varepsilon)$ vanishes outside the band, but the closed forms above are finite on a discarded root, so in code the test (A) must be applied explicitly. At the gap edge (for $|\varepsilon^*| \le W$, so that both branches are present) $\xi_\pm \to \xi^*$ and the two coherence-peak weights are in the ratio

$$\frac{N_t(+E_{\min})}{N_t(-E_{\min})} = \frac{1+\beta}{1-\beta}, \qquad \beta = \frac{\xi^*}{E_{\min}} = \operatorname{sgn}(\Delta_F)\,\frac{b}{\sqrt{1+b^2}} \tag{18}$$

This is, in closed form, the density-of-states-independent asymmetry between bias polarities that the paper points out after Eq. 37. For the example below $\beta = -0.316$, the ratio is 0.520, i.e. the $\eta = -1$ edge is 1.92 times stronger than the $\eta = +1$ edge.

7. Numerical illustration

Parameters as in the paper's Fig. 8 ($U = 90t$, $\Delta t = 4.5t$, $n = 1.875$, $k_B T = 0.01t$) with $V_{SO} = 0.5t$. Δ^0, Δ^s, μ and T_c come from my own solver of paper Eqs. 32, 33, 38 (`solve_gap.py`); everything else follows from the formulas above.

Δ^0/t	Δ^s/t	μ/t	k_BT_c/t	b	Δ_F/t	ε^*/t	Δ^*/t	$E_{\min}/t$
0.2926	1.3312	3.7265	0.484	0.3328	−0.9476	3.4426	−0.8531	0.8991

W/t	E_{+W}/t	E_{-W}/t	$\lvert\varepsilon^*\rvert \le W$?	ε_0/t **(node of Δ)**	E_0/t
4.2426	1.233	8.149	yes	0.879	2.85

Which branches exist where (this is the band test of Sec. 3 applied to the example):

energy range	**real branches**	$\Delta(E)$
$E < E_{\min} = 0.8991\,t$	none	undefined
$E_{\min} \le E \le E_{+W} = 1.233\,t$	$+$ and $-$	two-valued
$E_{+W} < E \le E_{-W} = 8.149\,t$	$-$ only ($\varepsilon_+ > W$ is discarded)	single-valued, $\Delta_-(E)$
$E > E_{-W}$	none ($\varepsilon_- < -W$ as well)	undefined

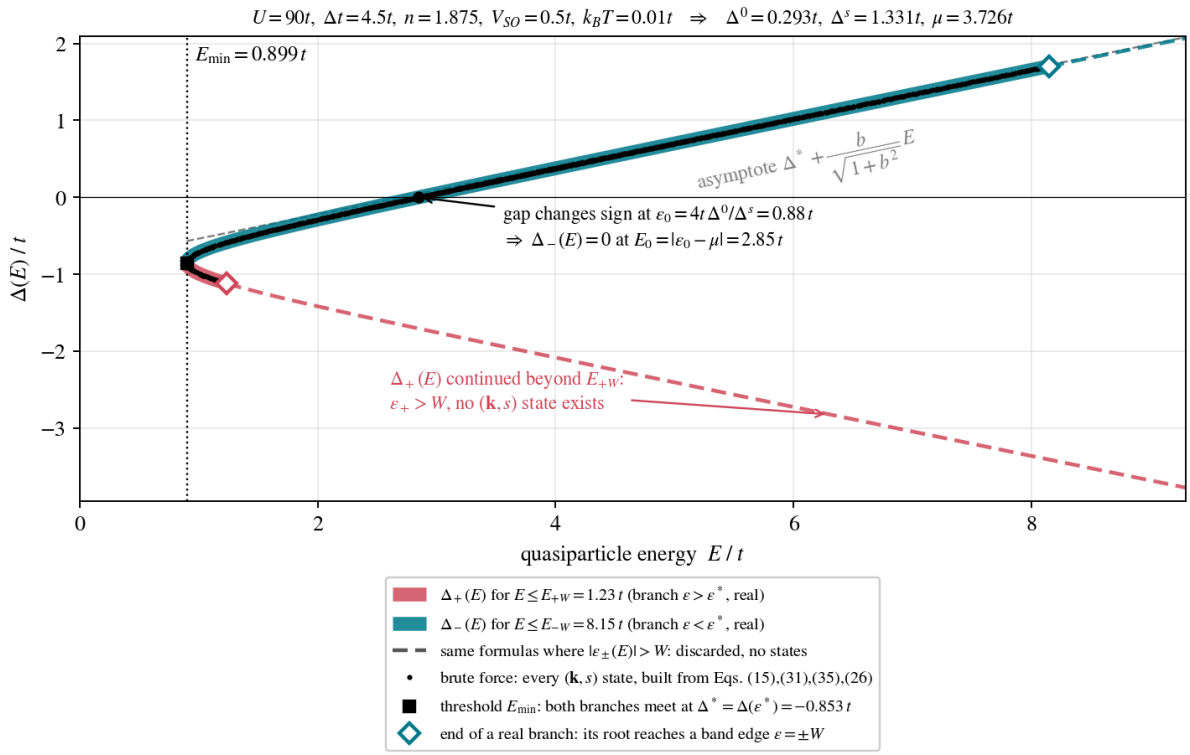


Fig. 3. $\Delta(E)$ for the example. Solid: the real branches Δ_+ (red, up to E_{+W}) and Δ_- (teal, up to E_{-W}) from (R). Black dots: $\bar{\Delta}_{\mathbf{k}s}$ against $E_{\mathbf{k}s}$ for every one of the 28,800 states, built directly from paper Eqs. 15, 31, 35 and 26 without any inversion. Dashed: the same formulas where the root lies outside the band — these continuations are discarded, no state exists there. Diamonds mark where a real branch ends; the branches meet at $(E_{\min}, \Delta^*)$.

Worked examples — each root is tested against $W = 4.2426\,t$ before Δ is accepted:

E/t	$+$ **branch** ($\varepsilon > \varepsilon^*$)			$-$ **branch** ($\varepsilon < \varepsilon^*$)		
	ε_+/t	$\lvert\varepsilon_+\rvert \le W$?	Δ_+/t	ε_-/t	$\lvert\varepsilon_-\rvert \le W$?	Δ_-/t
1.0	3.858	**kept**	−0.991	3.027	**kept**	−0.715
2.0	5.138	**discarded**	–	1.747	**kept**	−0.289
5.0	8.109	**discarded**	–	−1.224	**kept**	0.700
8.0	10.985	**discarded**	–	−4.100	**kept**	1.657
8.5	11.462	**discarded**	–	−4.577	**discarded**	–

At $E = 2t$ the $+$ root sits at $\varepsilon_+ = 5.138\,t > W$: it is discarded, and $\Delta(2t) = \Delta_-(2t) = -0.289\,t$ is single-valued. At $E = 8.5t$ both roots are outside the band: no state has that energy. Because the gap function changes sign at $\varepsilon_0 = 0.879\,t$ (inside the band), $\Delta_-(E)$ crosses zero at $E_0 = |\varepsilon_0 - \mu| = 2.85\,t$.

- **Inversion:** over all 28,800 $(\mathbf{k}, s)$ states $\max |\bar{\Delta}_{\mathbf{k}s} - \Delta_\pm(E_{\mathbf{k}s})| = 7{\times}10^{-15}\ t$ (using, for each state, the branch it belongs to); the smallest $E_{\mathbf{k}s}$ in the zone is $0.89914\,t$ against the formula value $0.89913\,t$ (paper Eq. 40).
- **By-products** (test function $F(E) = e^{-E/2}(1 + 0.3\cos E)$; direct state sum vs. the energy integrals with the band-masked branch sums; the small differences are histogram binning): N_s: 0.44651 vs 0.44646; $N_t(+E)$: 0.07996 vs 0.07994; $N_t(-E)$: 0.36655 vs 0.36652.

Cross-check of my solver against the paper's Fig. 8 (same $U, \Delta t, n, k_BT$; paper values read off the figure, so approximate). The largest deviations are 4% in T_c and 6% in $E_{\min}$.

V_{SO}/t	k_BT_c/t	**paper**	$E_{\min}/t$	**paper**	$2E_{\min}/k_BT_c$	b	β
0.05	0.329	0.33	0.591	0.57	3.59	0.267	−0.258
0.1	0.333	0.33	0.600	0.58	3.60	0.269	−0.259
0.2	0.352	0.35	0.635	0.62	3.61	0.277	−0.267
0.3	0.383	0.38	0.696	0.68	3.63	0.291	−0.279
0.4	0.427	0.42	0.783	0.75	3.67	0.309	−0.296
0.5	0.484	0.47	0.899	0.85	3.71	0.333	−0.316
0.6	0.553	0.53	1.044	1.03	3.77	0.360	−0.339

8. Minimal code and files

```
import numpy as np

def delta_of_E(E, D0, Ds, mu, VSO, t=1.0):
    """Physical branches of Delta(E): (Delta_plus, Delta_minus).
    NaN for E < Emin and where the root eps_pm lies outside the band (|eps| > W)."""
    E = np.asarray(E, float)
    b     = Ds / (4*t)
    DF    = D0 - b*mu                                # Delta at the Fermi level
    s     = np.sqrt(1 + b*b)
    W     = 4*t*np.sqrt(1 + VSO**2/(2*t**2))        # Rashba band:  -W <= eps_ks <= W
    Emin  = abs(DF)/s                                # paper Eq. 40
    estar = (mu + b*D0)/(1 + b*b)                    # paper Eq. 39
    Dstar = DF/(1 + b*b)                             # Delta(eps*)
    with np.errstate(invalid="ignore"):
        ell   = np.sqrt(E**2 - Emin**2)              # NaN below the threshold
        eps_p = estar + ell/s                        # band energy of the "+" root
        eps_m = estar - ell/s                        # band energy of the "-" root
        Dp = np.where(np.abs(eps_p) <= W, Dstar - b*ell/s, np.nan)   # band test (A)
        Dm = np.where(np.abs(eps_m) <= W, Dstar + b*ell/s, np.nan)   # discarded -> NaN
    return Dp, Dm
```

The listing was executed against the full module when this note was built: it reproduces the band-masked branches of `delta_of_E.py` on 4001 energies, NaN pattern included.

Files (in `/home/user/deltaE/`): `Delta_of_E_derivation.pdf` / `.html` (this note); `delta_of_E.py` (formulas, band test, brute-force check over all states, 300-set stress test of the band rule, by-product check, figures); `solve_gap.py` (self-consistent solver of paper Eqs. 32, 33, 38; prints the table above); `verify_symbolic.py` (sympy proof of Steps 1–5); `fig1_geometry.png`, `fig2_band_test.png`, `fig3_delta_of_E.png`; `build_doc.py` (rebuilds this note).